CCNA

VLAN and INTER-VLAN Routing TASK – 1

1. **Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).**

   Ans :

Feature	**LAN (Local Area Network)**	**WAN (Wide Area Network)**
Coverage	Covers a small area such as a home, office, or school.	Covers large geographical areas such as cities, countries, or the whole world.
Speed	Usually faster because devices are close together.	Generally slower than LAN because data travels over longer distances and multiple networks.
Ownership	Usually owned and managed by one person or organization.	Often uses infrastructure owned by multiple organizations or Internet service providers.
Daily-life example	**Zomato on home Wi-Fi:** Your phone connects to your home router through	**Instagram:** When you use Instagram, your phone communicates with Instagram's servers over

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
	a LAN before accessing Zomato.	the Internet, which is a WAN.

2. **Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data center in Bangalore using a WAN, labeling the LAN and WAN segments clearly.**

   Ans :

USER IN AHMEDABAD

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| LAN

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[Home Wi-Fi Router]

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======= WAN =======

| Internet / ISP |

| Ahmedabad → |

| Bangalore |

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[Flipkart Network]

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| LAN

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[Flipkart Data Center]

Bangalore

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3. **Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.**

Ans :

MPLS (Multiprotocol Label Switching) is a technology used to forward data efficiently across a network by attaching labels to packets.
In traditional IP routing, every router examines the destination IP address and performs a routing-table lookup at each hop.

With MPLS, the packet is classified and given a label at the network edge.
Routers in the MPLS core then use these labels to forward packets instead of repeatedly examining the IP header.
This makes forwarding simpler and can improve network efficiency and scalability.
MPLS also supports traffic engineering, allowing organizations to control routes and reduce congestion.
Therefore, MPLS provides faster, more predictable, and better-managed data routing than basic hop-by-hop IP routing.

4. **Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.

Hint: Look for how labels are used and how packets move between routers.**

   Ans :

   For the pre-configured MPLS network, you can note these two routing behaviors:

   1. **Labels are used for forwarding:** Unlike standard IP routing, where routers examine the destination IP address at every hop, MPLS routers use a **label** attached to the packet to decide where to forward it.
   2. **Packets follow a predefined label-switched path:** MPLS can send packets along a **Label Switched Path (LSP)** established through the network, rather than having each router independently choose the next hop using its IP routing table. This can provide more controlled and predictable packet movement.

5. **List three specific benefits MPLS offers over traditional routing for companies running high-traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real-world scenarios.**

   Ans :

   Three Benefits of MPLS

   1. **Better traffic management:** MPLS can use traffic engineering to direct heavy traffic along suitable paths. For services like **YouTube**, this can help reduce congestion when large numbers of users are streaming videos simultaneously.

2. **Predictable performance and QoS:** MPLS can prioritize important traffic, such as video or voice data.
   For **BookMyShow**, this can help keep ticket-booking transactions responsive during high-demand events such as major movie releases.
3. **Reliable and scalable connectivity:** MPLS provides controlled paths and supports fast rerouting when a network link fails. For a large service handling millions of users, this helps maintain connectivity and reduce service interruptions during network failures.